Can a model nobody trained on these cameras count vehicles for signal timing? Zero-shot CLIP on UVH-26

Yash Renwa · 2025AG05357 · AIML ZG525 Computer Vision, EC-1 Assignment 2 · Track C (vision-language, zero-shot classification) · Code and notebook: github.com/YashRenwa/uvh26-clip-zeroshot

1. Problem and track

Green time at a junction is set from approach demand in Passenger Car Units (PCU, IRC:106-1990): a truck or bus weighs 3.28 cars, an auto-rickshaw 2.0, a two-wheeler 0.75. Assignment 1 met this need with a detector fine-tuned on labelled frames. A new camera has no labels. This project asks whether a frozen vision-language model can name each vehicle from words alone, well enough for PCU, and whether its own score can say which vehicles a person must check. Boxes are the dataset's ground-truth boxes, so only classification is measured; per-frame PCU is the sum of the factors of the predicted IRC categories.

> **[YASH] Why Track C, and not A or B** (write in your own words; notes):
> - The constraint is the camera, not the model: zero labelled frames at deployment. Only a zero-shot vision-language model gives class names without training data; classes are described in text.
> - Track A (ViT vs CNN) compares two models that both need a labelled training set; at an unlabelled camera neither can be trained, and A1 already measured what labelled training buys.
> - Track B (generation) produces images, not counts; synthetic traffic would not estimate demand at this junction.
> - Track C lets the measurement be the point: how good is a frozen model here, and can its score route work to a person.

2. Data, licence and contamination

Source. UVH-26 (IISc AIM with Bengaluru Traffic Police; **CC BY 4.0**; Hub revision 59f82c5): 1920×1080 frames from Bengaluru Safe City cameras, captured February 2025, 14 classes, STAPLE (ST) consensus boxes. **Split.** Assignment 1's pseudo-camera split, reused unchanged and checksum-verified (sha256 f8181d24…): val 452 images (355 pseudo-cameras), test 477 (410), 0 pseudo-cameras in both. After dropping 51 exact duplicate boxes per split, val has 6,630 crops and test 6,299. Each crop is a square around the box with 10% context, padded with CLIP's mean pixel. Mini-bus, Tempo-traveller and Others have fewer than 30 test crops and are left out of the headline macro-F1.

What the model has seen. CLIP ViT-B/16 was trained on 400M web image–text pairs collected before 2021; these frames were captured in 2025 and published after, so the test images themselves cannot be in its training data. Captioned Indian street photos ("auto-rickshaw", "Tata Ace") very likely are, which is what zero-shot relies on. **Within the project**, prompts and the review threshold were tuned on val only. A by-eye audit of 40 seeded random test images against their most similar val image found 4 same-camera pairs: val–test camera overlap ≈ 10% (95% CI 4%–23%). Gradient-thumbnail similarity misses some of them (Figure 2), so the audit, not a threshold count, is the estimate. Prompts were written from the class definitions and revised at most twice, so this overlap can flatter test only slightly.



Figure 1. One seeded random val crop per class, as CLIP sees it (faces blurred). The Bicycle example is a car carrying a duplicate STAPLE box labelled Bicycle; see the label-noise finding in Step 7.



Figure 2. Leak audit pair #12: a test frame (left) and its nearest val frame (right) are the same camera (same parked red hatchback and grey SUV, same pillar sign), yet their gradient similarity is only g = 0.25 because a different truck fills each frame.

3. The frozen baseline

`openai/clip-vit-base-patch16`, revision 57c2164 (pinned in code and printed in the notebook), fp32 on CPU, used as published: CLIP's own processor, L2-normalised embeddings, logits = CLIP's logit scale (100) × cosine, softmax over the 14 class prompts, prediction = argmax. Nothing is fine-tuned, no temperature is fitted. Baseline prompt: "*a photo of a {class name}.*". For PCU, the IRC category is the one with the highest **summed** softmax over its classes, and that sum is the IRC score; this rule was fixed on val before test. The argmax→category version is reported as a secondary column.

4. One controlled intervention: the text prompts

The ensemble arm changes only the text: per class, it averages the text embeddings of six camera-aware templates ("a CCTV image of {}", "a top-down traffic camera view of {}", …) × several descriptive names written from the dataset card's class table (e.g. Three-wheeler: "an auto-rickshaw", "a green and yellow auto-rickshaw"; Others: "an agricultural tractor", "a road roller"). Image embeddings are computed once and shared by both arms; the notebook asserts their hash is identical. Design used val only, with at most two revisions, and the baseline's val confusion was already known when the first version was written. The version with the best val IRC accuracy was frozen by sha256 before test was opened:

Val (6,630 crops)	baseline	v1	v2 (frozen)	v3
IRC accuracy	0.586	0.731	**0.749**	0.748
PCU MAE per frame (argmax class)	8.43	2.61	**2.27**	2.27

5. Evaluation on data never tuned on

Test was opened only after the prompts and threshold were frozen. The notebook's fresh-kernel run re-scores it with the same files and reproduces the first opening's image embeddings exactly. Intervals are 95% paired percentile bootstrap over pseudo-camera groups (1,000 resamples, seed 42).

	Val baseline	Val ensemble	Test baseline	Test ensemble	Test difference (95% CI)
14-class accuracy	0.473	0.628	0.455 [0.435, 0.476]	**0.626** [0.610, 0.644]	+0.157 to +0.182
Balanced accuracy	0.222	0.336	0.205	**0.378**	
Macro-F1 (11 classes ≥ 30 crops)	0.249	0.388	0.230 [0.214, 0.241]	**0.396** [0.377, 0.416]	+0.147 to +0.187
IRC accuracy (summed)	0.586	0.749	0.575 [0.559, 0.592]	**0.742** [0.729, 0.755]	+0.154 to +0.179
IRC accuracy, argmax class (secondary)	0.533	0.745	0.514	0.739	
PCU MAE per frame	4.81	2.08	4.77 [4.30, 5.19]	**1.98** [1.79, 2.17]	–3.20 to –2.33
as share of mean demand; bias	26.0%; +4.19	11.2%; +0.29	28.9%; +4.10	12.0%; +0.72	
PCU MAE, argmax class (secondary)	8.43	2.27	8.55	2.22	

Table 1. Both arms, same image embeddings, val and test. Test: 6,299 crops, 477 frames, mean demand 16.48 PCU per frame. Headline arm (chosen on val): ensemble.

Finding: the prompt change alone is a large, robust gain. On unseen cameras the ensemble names the IRC category of 74.2% of vehicles against 57.5%, and cuts the per-frame PCU error from 4.77 to 1.98 (12.0% of demand); every bootstrap resample favours it. The baseline over-counts demand (bias +4.10 PCU per frame), so its errors push vehicles into heavier categories. Val and test agree closely for both arms, so tuning the prompts on val did not over-fit it.

Where accuracy is lost (Figure 3). Cars, two-wheelers and autos are mostly right. Trucks and buses are the weak category, mostly called LCVs or autos, and LCVs are often called cars. Car subtypes are confused with each other (14-class accuracy 62.6%), which costs nothing in PCU.

What labelled training bought (reference, not a baseline): Assignment 1's fine-tuned YOLOv8 on the same test frames reached PCU MAE 2.61 per frame (15.8% of demand). That figure includes its detection errors and ours does not, so the two are not directly comparable.

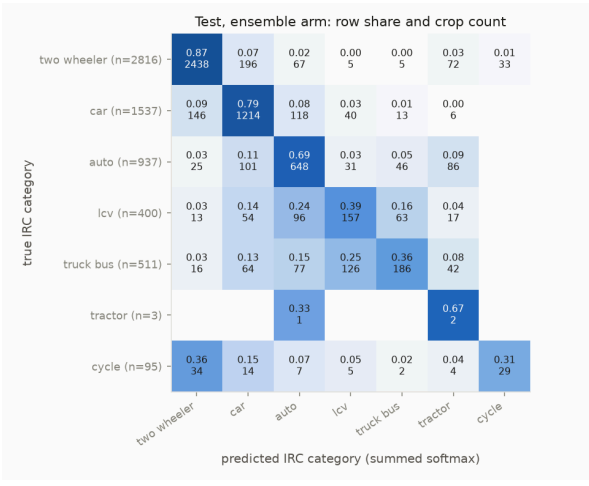


Figure 3. Test IRC-category confusion, ensemble arm (row share and crop count).

6. Confidence against correctness

The score is a softmax value (14-class) or its sum over a category (IRC). It is not a probability: whether it tracks correctness is measured, not assumed. With the ensemble prompts it does track correctness (Figure 4): test ECE of the IRC score is 0.020 (0.023 with equal-mass bins), against 0.051 for the baseline. The coverage curve shows accuracy rising steadily as the threshold rises.

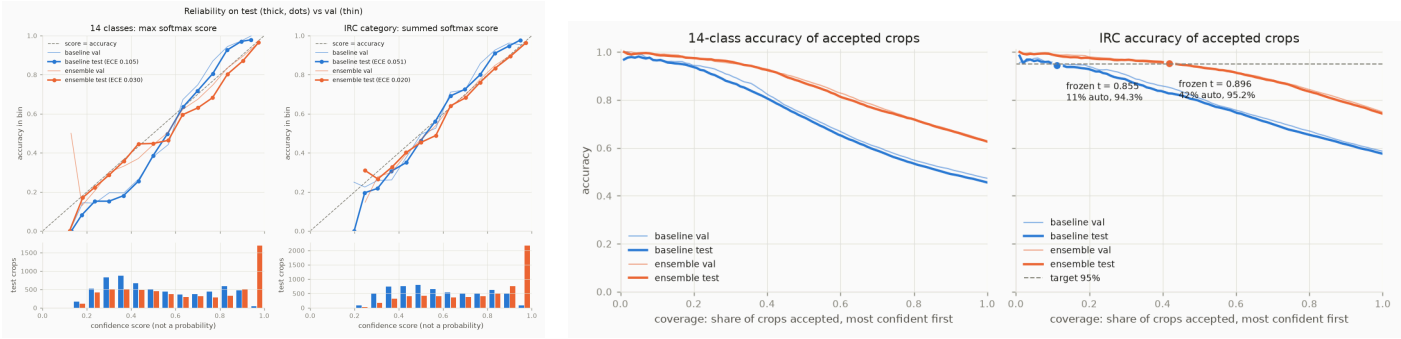


Figure 4. Left: reliability diagrams, 15 equal-width bins (14-class score and IRC score). Right: coverage curves, accuracy of the crops kept vs share kept. Thick: test; thin: val.

The review rule. A crop is accepted without review if its IRC score ≥ *t*; *t* is the lowest threshold at which accepted val crops reach 95% IRC accuracy (target set before test), frozen on val for both arms.

Review rule on test	*t* (val)	Accepted unseen	Accepted IRC accuracy (95% CI)	Accepted 14-class acc.	Sent to a person	PCU MAE before → after review
Baseline	0.8546	11.2%	94.3% [92.5%, 96.0%]	91.9%	88.8%	4.77 → 0.08
Ensemble	0.8956	**41.8%**	**95.2%** [94.2%, 96.2%]	83.6%	58.2%	1.98 → 0.24

The ensemble rule held its target on unseen cameras: 41.8% of crops (val 40.8%) need no person, at 95.2% IRC accuracy. The baseline's rule was fragile on val (its coverage curve hovers near 95% at low coverage) and on test accepts only 11.2%, below target. The rule targets IRC categories, so accepted crops are right at the 14-class level only 83.6% of the time; per-class counts of car subtypes are not promised. **95.2% is a lower bound:** in the seeded draw of 10 accepted errors, 5 are label errors where CLIP is right (Section 7).

7. Failure analysis

Draw. With the project seed, 30 errors were drawn at random from all 2,356 14-class test errors of the ensemble, then 10 from the 126 errors the review rule lets through (IRC score ≥ *t*, IRC wrong). None was chosen by hand. Each was tagged with one cause from the crop and its frame (tags judged by Claude at my request and reviewed by me).

Cause	Random 30	Accepted 10
Occlusion by another vehicle	7	1
Small (long side < 96 px)	6	1
Look-alike class	5	0
A neighbour fills the crop	4	2
Cut by the frame edge	3	1
IR / night frame	2	0
Label error (CLIP right)	2	**5**
Other	1	0

All 6,299 test crops	Crops	IRC error rate
Overlap with another box: IoU 0	1,686	22.1%
IoU ≥ 0.3	919	31.9%
Inside the frame	5,286	23.8%
within 3 px of the frame edge	1,013	36.2%
Long side ≥ 96 px	4,796	23.8%
32–95 px	1,496	31.9%
True class Truck	339	78.2%



Figure 5. The first five of the seeded random draw (not chosen by hand), with true and predicted class, score and cause tag. Faces blurred.

Occlusion, neighbouring vehicles, frame-edge cuts and small size account for 20 of the 30 random errors, and each tag raises the error rate across all test crops, not just the drawn ones (second table). **Trucks are the costliest class for PCU:** 78.2% of trucks get the wrong IRC category. They are 5.4% of crops but carry 23.8% of all per-crop PCU error (434 of 1,823 PCU), almost all of it under-count (net –378): a truck (3.28) read as an LCV (1.54), auto (2.0) or car (1.0). The review rule does not catch them: 27 of the 29 trucks it accepts are wrong. This weakness was already visible on val before test was opened.

[YASH] The named failure pattern (your wording; notes):

- Candidate: "CLIP names what fills the crop, not the vehicle in the box: occlusion, neighbours, frame-edge cuts and small size account for 20 of 30 random errors."
- Support: error rate rises with overlap (22.1% → 31.9% IRC; 14-class 30.2% → 57.7%), at the frame edge (36.2% vs 23.8%) and for small crops (31.9% vs 23.8%).
- Name trucks as the costliest class for PCU, and say why (a large vehicle partly in view or behind others reads as an LCV or auto).

Label noise. Some vehicles carry two ST boxes with different classes at IoU ≥ 0.99 (STAPLE double labels): 279 groups, 568 test crops (9.0%); Figure 1's Bicycle is one. As a secondary check defined after test was opened (headline unchanged), excluding these crops moves the ensemble to accuracy 0.665, IRC accuracy 0.753, PCU MAE 1.82 and accepted IRC accuracy 95.9%. Assignment 1's labels carry the same duplicates.

8. Ethics and privacy

Risk in this system. CLIP matches images against any text. The same pipeline on city cameras could be asked for a person ("a woman in a red saree riding a scooter") instead of a vehicle class, turning a counting tool into a search tool for individuals; the frames also show faces and number plates. **Mitigations implemented.** (1) Closed vocabulary: classifiers are built only from prompt files registered by sha256, and the notebook shows a free-text prompt file being refused. (2) Image embeddings stay in memory in the notebook run and no embedding file is written (asserted); outputs are per-crop class, score and per-frame counts. (3) Every figure with pixels passes a face blur (OpenCV Haar) plus manual boxes, and each was checked by eye. **Costed, not built:** a number-plate detector (one more model with its own misses).

[YASH] Ethics judgement (your view; notes):

- Is the closed vocabulary enough? It stops misuse of this code, not of CLIP; anyone with the weights can type free text.
- Who reviews the 58% of crops sent to a person, and what they see (crops, not whole frames).
- Error is not neutral: under-counted trucks shorten green time on freight-heavy approaches.

Conclusion

[YASH] Conclusion (your call; notes):

- What you would deploy: the ensemble prompts with the review rule (t = 0.8956): 42% of crops need no person at ≥ 95.2% IRC accuracy; PCU MAE 1.98 without review.
- The number you refuse to promise: truck/bus counts, and car-subtype counts.
- One change was enough to matter: prompts only, same image embeddings.
- Caveats: ≈10% val–test camera overlap; ground-truth boxes (no detection errors); label noise makes accepted accuracy a lower bound.

Declarations and references

AI assistance. Claude (Anthropic), working through Claude Code, wrote the code in the repository, ran it on my machine, drafted the prompt files from the dataset card under my constraints, proposed analysis options, judged the leak-audit pairs, the 40 failure-cause tags and the remaining face detections by eye at my request (I reviewed them), and drafted the factual sections of this report from `report/numbers.json`. I chose the problem, the track and the reuse of Assignment 1's split, and made the decisions recorded in the decision log: the 95% IRC review target, the redaction method, the leak audit and its reported estimate, the prompt constraints and selection rule, the summed-softmax IRC rule, the frozen prompts, threshold and headline arm, the duplicate-label check, and the cause tags. Sections marked [YASH] are mine. No hosted API was used; no data left my machine except downloads from the Hugging Face Hub.

Dataset. A. Sharma et al. *Towards Image Annotations and Accurate Vision Models for Indian Traffic, UVH-26-v1.0.* IISc, 2025. arXiv:2511.02563. CC BY 4.0. · A. Radford et al. Learning transferable visual models from natural language supervision (CLIP). ICML 2021. · S. K. Warfield, K. H. Zou, W. M. Wells. STAPLE. IEEE TMI 23(7), 2004. · Indian Roads Congress. IRC:106-1990.